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Indirect bioleaching for metal recovery from battery black mass using microbially generated acid from a secondary sulfur resource

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Recycling critical metals from spent lithium-ion batteries (LIB) has attracted significant attention in recent years, driving the search for more sustainable and environmentally friendly recovery methods. Among emerging approaches, indirect bioleaching may be a very promising alternative for recovering valuable metals from spent LIB. In this study, sulfuric acid (H₂SO₄) was produced by using a mixed culture containing *Acidithiobacillus thiooxidans* and *Acidithiobacillus caldus*, supplemented with elemental sulfur recovered as a waste product from a biogas plant. The biogenic acid was subsequently used for leaching black mass derived from nickel manganese cobalt (NMC)-based lithium-ion batteries (LIB). First, the influence of different concentrations (0.2, 1, and 2 M) of hydrogen peroxide (H₂O₂) on metal dissolution was investigated. Subsequently, 2% (v/v) of rhamnolipids / phospholipids was added and the effect on the H₂SO₄–H₂O₂ system was studied. Finally, the best performing system was scaled up to a 1L reactor, achieving complete recovery of Mn and Li from 10% (w/v) NMC within 1 h. This result highlights the high potential of microbially produced H₂SO₄ as an effective leaching agent for recovering critical metals from spent LIB, while simultaneously valorizing a secondary sulfur waste stream, aligning with the broader objectives of circular and sustainable resource recovery approaches.

KEYWORDS

biogenic acid, bioleaching, biosurfactants, critical metals, hydrogen peroxide, lithium-ion batteries, recycling

1 Introduction

The increasing demand for lithium-ion batteries (LIB) has been driven by their widespread use in electric vehicles (EVs), renewable energy storage systems, and consumer electronics. For instance, the number of EVs is expected to increase to 965 million in 2050 (Llomas-Orozco et al., 2023). Spent LIB pose significant environmental risks due to their toxic and flammable components, which can contaminate the environment if not properly managed. At the same time, LIB present a valuable secondary resource of critical and

strategic raw materials such as Li, Co, Ni, and Mn. Therefore, efficient recycling of LIB could be essential to close material loops, mitigate environmental risks, and reduce primary mining of these materials (Zanoletti et al., 2024).

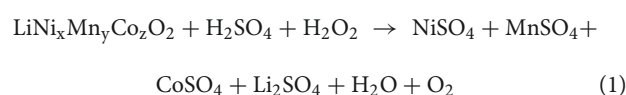
LIB recycling is currently performed mainly using pyrometallurgy and hydrometallurgy methods. Although these methods have high recovery efficiencies, they are associated with significant drawbacks including high energy inputs, use of harsh chemicals such as concentrated acids, and generation of secondary waste streams (Liu et al., 2024). In this context, bioleaching presents a promising alternative. Bioleaching is a biotechnological process that uses microorganisms or their metabolites to solubilize metals. Conventional bioleaching, which has been extensively applied to process low-grade sulphidic ores and gold concentrates, uses a direct approach that partially relies on the solubilization of valuable metals through direct microbial interaction with the solid substrates, often assisted by biofilm and exopolysaccharide layer (EPS) formation (Vera et al., 2013). However, the application of direct bioleaching to extract metals from complex materials such as LIB often faces inhibitions of microbial growth and metabolic activity by toxic metals, electrolytes, binders, and other components present in the waste stream. Furthermore, direct bioleaching is constrained by challenges associated with high pulp densities and often require external pH adjustments to maintain conditions favorable for microbial growth and activity (Kiskira et al., 2026). To overcome this limitation, indirect leaching is applied to separate the microbial generation of the leaching agents (i.e., organic/inorganic acids and/or Fe^{3+}) from the subsequent recovery of valuable materials from LIB (Boxall et al., 2018; Pakostova et al., 2024). Increasing research attention has been directed toward the bioleaching of LIB with a range of innovation approaches being developed to improve metal recovery. Naseri and Mousavi (2024) reported that a step-wise addition of biogenic acid produced by *Acidithiobacillus* (*A.*) *thiooxidans* during bioleaching of spent LIB increased the leaching efficiency of Mn and Li by 50% and 38%, respectively, and reduced the leaching duration from 16 to 8 days, when compared with direct bioleaching. Kim et al. (2024) reported that an application of static magnetic field enhanced the leaching efficiency of Li during bioleaching of spent LIB using *A. ferrooxidans*, a magnetotactic bacterium. High dissolution of Li (1.6 g/L) was achieved during bioleaching of LFP even at a high pulp density of 125 g/L and 89% of Li was recovered as Li_3PO_4 from the leachate solution by precipitation (Gu et al., 2025). The same study also determined that the bioleaching process has a lower carbon footprint, generating an average of 0.98 kg/ $\text{CO}_{2\text{eq}}$, compared with other battery recycling methods.

Sulfuric acid-mediated leaching is widely regarded as an effective approach for metal recovery from LIB black mass, offering high extraction efficiencies under relatively mild operating conditions (Lee et al., 2025). Biogenic sulfuric acid produced via microbial oxidation of reduced sulfur compounds represents a promising alternative to industrially produced sulfuric acid, as using biogenic acids reduces the carbon footprint and environmental hazards associated with chemical acid production. Successful applications have been reported by our group and others for valorization of metals from different waste streams such as waste incineration residues and pharmaceutical blisters (Kremser et al., 2022, 2023). Indirect leaching of LIB using biogenic sulfuric acid has been reported (Boxall et al., 2018; Pakostova et al., 2024), but

it remains underexplored, particularly in relation to the role of chemical and biological additives in improving metal dissolution kinetics and process efficiency.

LIB encompasses a variety of cathode chemistries such as lithium nickel manganese cobalt oxide (NMC), lithium iron phosphate (LFP), and lithium cobalt oxide, with each type optimized for specific performance requirements (Manthiram, 2020). Although Co-free cathode materials such as LFP have recently been introduced, NMC has historically dominated the battery market predominantly used in EVs due to their high energy density (Fallah and Fitzpatrick, 2023). This means that NMC will constitute the largest fraction of LIB waste in the near future. In addition, NMC cathodes contain significant quantities of critical metals such as Li, Ni, Co, and Mn, making them attractive targets for resource recovery.

In NMC-based cathode materials, transition metals are present as mixed metal oxides ($\text{LiNi}_x\text{Mn}_y\text{Co}_z\text{O}_2$). Efficient dissolution of NMC requires acidic conditions, ideally in combination with a reducing agent or catalytic pathway, owing to the inherently high oxidation states and structural stability of the metal oxides (Meshram et al., 2016). In NMC, the transition metals are commonly present in their thermodynamically stable higher oxidation forms, with Mn mainly as Mn^{4+} , Co as Co^{3+} , and Ni as a mixture of $\text{Ni}^{2+}/\text{Ni}^{3+}$ (and Ni^{4+} in charged states). These species exhibit limited solubility under acid-driven leaching alone. Reducing agents such as hydrogen peroxide (H_2O_2) facilitate the reduction of Mn^{4+} and Co^{3+} to the more soluble Mn^{2+} and Co^{2+} (while also assisting in the reduction of higher-valent Ni species), accelerating thus overall metal dissolution kinetics. In acidic media, H_2O_2 is oxidized to molecular oxygen, acting as a reducing agent for metal ions such as Co^{3+} (Lee and Rhee, 2003). As a result, introducing a reducing agent or catalytic species is required to facilitate redox reactions that convert these metals into more soluble oxidation states, thereby improving overall leaching kinetics. Importantly, the concentration of H_2O_2 is critical for efficient metal leaching; insufficient H_2O_2 limits the reduction of metal ions in higher oxidation states, whereas excess H_2O_2 may decompose to water and oxygen, reducing redox efficiency and potentially altering the solution potential. Optimization of H_2O_2 dosing is thus essential to maximize metal extraction. Equation 1 schematically represents the acid- and peroxide-assisted leaching of NMC material (Vieceli et al., 2023).



In addition to reducing agents, surfactants can influence leaching kinetics and interfacial reactions by modifying surface chemistry and enhancing mass transfer. Commercial surfactants are predominantly petroleum derived and therefore present potential environmental risks. On the other hand, biosurfactants are produced by microorganisms and are biodegradable, making them more environmentally friendly and less harmful (Mishra et al., 2021). Biosurfactants such as rhamnolipids and phospholipids possess amphiphilic structures that enable adsorption onto solid surfaces, potentially increasing wettability and promoting particle dispersion. These effects can enhance contact between the acidic leach liquor and the substrate surface, thereby facilitating metal dissolution and associated redox

reactions at the solid–liquid interface (Roy, 2018). Additionally, biosurfactants may form weak complexes with dissolved metal ions, influencing metal speciation and transport in solution (Chernyshova et al., 2023). Biosurfactants have been used for different applications such as removal of metals from contaminated sites (Mulligan et al., 2001; Diaz et al., 2015; Mishra et al., 2021). Despite their demonstrated effectiveness in bioleaching and metal remediation systems, the role of biosurfactants in LIB black mass leaching has not yet been systematically examined.

In this study, microbially generated sulfuric acid produced from waste sulfur was evaluated for leaching valuable metals from NMC-based black mass. The effect of varying H_2O_2 concentrations on the extraction efficiencies of Ni, Mn, Co, and Li was systematically investigated to determine the optimal reducing agent dosage. Subsequently, two biosurfactants, rhamnolipids and phospholipids, were introduced into the bio-generated H_2SO_4 system under the optimized H_2O_2 conditions to assess their influence on metal dissolution. The best-performing system was then applied in a scale-up (1 L) leaching experiment at high pulp density and room temperature. To the best of the authors' knowledge, the combined use of biogenic acid derived from waste sulfur with a systematic evaluation of H_2O_2 and biosurfactants for the leaching of black mass under ambient conditions and without external heating has not previously been reported. This integrated approach demonstrates a low-energy route for LIB recycling and contributes to the development of more sustainable metal recovery processes.

2 Materials and methods

2.1 NMC-based active cathode material

Black mass (BM) derived from lithium nickel manganese Cobalt Oxide (NMC) cathode material was obtained from a commercial recycling company in Germany. Metals contents in the BM were determined by inductively coupled plasma mass spectrometry (ICP-MS), in accordance with ÖNORM EN 13657:2002-12 (Table 1). In addition to the elements listed in Table 1, the NMC based BM consisted of C, which constitute a majority of the BM and traces of other elements.

2.2 Biogenic H_2SO_4 production

A mixed culture of *A. thiooxidans* DSM 1927 and *A. caldus* DSM 9466 (1:1) both purchased from the German Collection of Microorganisms and Cell Cultures GmbH (DSMZ) were first cultivated separately, each in a 250 mL Erlenmeyer flask for 21 days at 30 °C and 150 rpm in an Infors HT Multitron Pro shaker (Bartelt, Austria). For the biogenic acid production, the two grown cultures were mixed (1:1) (v/v) and inoculated into a

2 L reactor (Schott DURAN, Germany) with a working volume of 1.8 L using basalt salt medium (pH 3.5) supplemented with 1% (w/v) sulfur. The basalt medium was prepared as described by Nancucheo et al. (2016). The sulfur in a form of a dried powdered product was obtained from the desulfurization unit of a biogas plant in Denmark. The reactor was maintained at 30 °C and agitated at 250 rpm using a magnetic stirring plate (IKA RCT Digital, Germany) connected to a pH monitor and 50 L/h supply of pressurized air. After 14–21 days of cultivation, 1.2 L of the solution was removed and replaced with a fresh medium and S, with the remaining 0.6 L serving as an inoculum for the new batch. The collected solution was then centrifuged at 9,000 rpm for 20 min (SORVALL LYNX 6000, Thermo Scientific, Germany) to remove the solid residue and the biogenic acid was then stored at 4 °C. The acid production was carried out over a total of 10 cycles, yielding approximately 12 L of biogenic H_2SO_4 solution. The accumulated acid was subsequently used directly, without any dilution, adjustment, or other modification, in all leaching experiments reported in this paper. The acid concentration was determined by ion chromatography using Dionex ICS-900 (Thermo Fisher Scientific, United States) and confirmed by acid-base titration using phenolphthalein as indicator (Fluka Analytical) and 0.1 M NaOH as the titrant.

2.3 Indirect bioleaching

Batch indirect bioleaching was performed in 100 mL Erlenmeyer flasks (50 mL working volume) containing biogenic acid and 10% (w/v) NMC, placed on a magnetic stirring plate (VARIOMAG Poly, Thermo Scientific, Germany). All experiments were carried out in duplicate for 2 h at room temperature (22 °C) and 250 rpm, with samples being withdrawn every 20 min for further analyses. Different concentrations of H_2O_2 (0.2/1/2 M) were tested to investigate the effects of the reducing agent on the metal leaching efficiency. Further, the effects of two biosurfactants, namely 2% (v/v) rhamnolipids (RL) (AGAE Technologies, USA) and 2% (v/v) phospholipids (PL) (Soy Phospholipids, Sigma-Aldrich), were tested in combination with the optimal concentration of H_2O_2 . For this, 10% NMC was added to the H_2SO_4 - H_2O_2 solution and the biosurfactants (PL/RL) were added after 40 min, after which the experiment continued for 2 h.

Finally, the leaching system exhibiting the best performance was scaled up to a 1 L reactor comprising a jacketed reaction vessel with a flat bottom (DN 100, Schmizo, Switzerland), equipped with 2-blade impeller connected to a mechanical stirrer (Heidolph RZR 2021, Germany) and a condenser to prevent liquid loss due to evaporation. Sampling was performed at regular intervals in the first 2 h, and the experiment further continued for 24 h. The scaled leaching experiment was carried out in duplicate at room temperature and 250 rpm.

2.4 Analytical methods

Oxidation reduction potential (ORP) and pH measurements were regularly conducted using Inlab Redox (vs. Ag/AgCl) and LE422 electrodes respectively, both connected to a S220 pH/ion meter (Mettler Toledo, Switzerland). Liquid samples

TABLE 1 Elemental composition of NMC-based black mass.

Elemental concentration (wt %)						
Ni	Mn	Co	Li	Cu	Al	Fe
5.86	4.17	14.50	2.76	3.50	5.26	0.57

collected during leaching experiments were filtered using 0.2 μm polyamide syringe filters (Macherey-Nagel, Germany) prior to metal concentration determination by inductively coupled plasma – optical emission spectrometry (ICP-OES), using a 5110 ICP-OES connected to a SPS4 Autosampler (both Agilent Technologies, United States). The leaching efficiency of metals were calculated using Equation 2. The post-leaching solid residues were analyzed using a TM 3030 scanning electron microscope (SEM) equipped with an energy-dispersive X-ray spectroscopy (EDS) detector (Hitachi, Japan).

$$\text{Leaching efficiency (\%)} = [(C_L - C_B) / C_{\text{NMC}}] \times 100 \quad (2)$$

where, C is metal concentration in leachate solution (C_L), biogenic acid (C_B) and NMC black mass (C_{NMC})

3 Results and discussion

3.1 Biogenic H_2SO_4

Biogenic sulfuric acid produced by a co-culture of *A. thiooxidans* and *A. caldus* cultivated in the presence of biogas plant derived sulfur reached a concentration of 0.63 M and a pH of 0.44 and was subsequently used in the indirect leaching experiments. Kremser et al. (2023) reported that the elemental sulfur derived from a biogas plant exhibited faster oxidation by a mixed culture of *A. thiooxidans* and *A. caldus* compared with commercial elemental sulfur. The same study also demonstrated improved Al leaching performance using the biogenic acid system. The authors attributed this enhanced performance to the presence of bioactive compounds in the bio-sulfur (Kremser et al., 2023). In a study by Valimets et al. (2026), biogas derived elemental sulfur was found to be more effective substrate than commercial sulfur for microbial sulfuric acid production. This was ascribed to its higher water solubility and the presence of organic compounds, such as carbohydrates and proteins, in the biogas-derived sulfur. Based on these findings, biogas-derived elemental sulfur was selected for the biogenic acid production, which was subsequently used in the leaching experiments conducted in this study.

3.2 Indirect leaching of NMC

During indirect leaching of 10% NMC using biogenic H_2SO_4 supplemented with varying concentrations of H_2O_2 , the addition of H_2O_2 generated strongly reducing conditions, as evidenced by the decrease in ORP. These conditions promoted the dissolution of lattice-bound Co^{3+} and higher-valent Ni species. At the same time, a gradual increase in pH was observed in H_2O_2 -containing flasks (Figures 1A,B), reflecting proton consumption during reductive dissolution reactions and partial H_2O_2 decomposition. These changes were associated with improved extraction of Co (Figure 1C) and Ni (Figure 1D) relative to the H_2O_2 -free control, confirming that acid dissolution alone is insufficient to efficiently disrupt the layered oxide structure of NMC materials. Of note, for Ni, moderate but consistent increase in recovery was seen upon H_2O_2 addition, which is consistent with the presence of

reducible higher-valent Ni species within the cathode structure. In contrast, Li recovery remained consistently high (>80%) across all treatments, including the control, indicating that its dissolution is primarily governed by proton attack rather than redox processes, as previously reported (Xin et al., 2009).

Notably, increasing the H_2O_2 concentration above 0.2 M did not result in proportional improvements in metal recovery. This suggests that excess H_2O_2 undergoes rapid decomposition, reducing redox efficiency and potentially promoting secondary effects such as pH-dependent precipitation of dissolved species. H_2O_2 exhibits both oxidizing and reducing behavior, depending on the electrochemical environment. Variations in the ORP together with an excess concentration of H_2O_2 can shift its role from a reducing agent to an oxidizing agent. Such condition may promote the re-oxidation of certain divalent, resulting in their precipitation and the formation of surface passivation layers, thereby hindering further metal dissolution (Miao et al., 2026). As 0.2 M H_2O_2 achieved metal recoveries comparable to those obtained with 1 and 2 M H_2O_2 , higher concentrations were considered unnecessary. Similar observations were reported by Baniasadi et al. (2025), who investigated NMC leaching with biogenic acid containing different concentrations of H_2O_2 (0–5% v/v). In that study, 0.5% H_2O_2 was the optimal concentration, while increasing H_2O_2 concentrations led to a loss of Co and Ni at the beginning of the leaching (Baniasadi et al., 2025). Therefore, 0.2 M H_2O_2 was selected for all subsequent leaching experiments.

To investigate the effect of biosurfactants during BM leaching using biogenic acid and 0.2 M H_2O_2 , 2% RL or 2% PL were added after 40 min of leaching, when the pH had exceeded 4 (Figure 2A), as biosurfactants generally exhibit greater stability at near-neutral pH (Abdel-Mawgoud et al., 2009). The addition of either 2% RL or 2% PL did not enhance metal solubilization despite the presence of 0.2 M H_2O_2 , and instead, resulted in slightly lower recoveries of Co, Ni, and Mn compared with the control (H_2SO_4 - H_2O_2 system) (Figures 2C–F). Furthermore, biosurfactant addition raised the solution pH, particularly in the PL-amended system, and partially changed the ORP profile shown in Figures 2A,B. These changes have likely affected the redox conditions required for efficient dissolution for transition metals from the NMC matrix. Biosurfactants possess a diverse array of functional groups, including hydroxyl, carboxyl, amino, carbonyl, and fatty acid moieties. These functional groups exhibit strong affinity to metal ions through multiple interaction mechanisms, including ion exchange, chelation, and electrostatic attraction (Yang et al., 2021). The changes in pH and ORP after the addition of PL and RL can be attributed to the ionizable functional groups present in the biosurfactant molecules and their interaction with dissolved metal species.

A recent study by Chakankar et al. (2026) investigated the use of rhamnolipids as floatation agents for metal recovery from a battery recycling process water and reported that the metal-binding affinity of rhamnolipid followed the order $\text{Mn}^{2+} > \text{Ni}^{2+} > \text{Co}^{2+}$, whereas Li^+ showed negligible interaction with the biosurfactant. These findings are consistent with the observations in the present study, where the addition of biosurfactants to the H_2SO_4 - H_2O_2 system reduced the dissolution of Mn, Ni, and Co, while Li recovery remained largely unaffected (Figures 2C–F). The above findings suggest that interactions between dissolved metal ions and biosurfactants may lead to the formation of insoluble

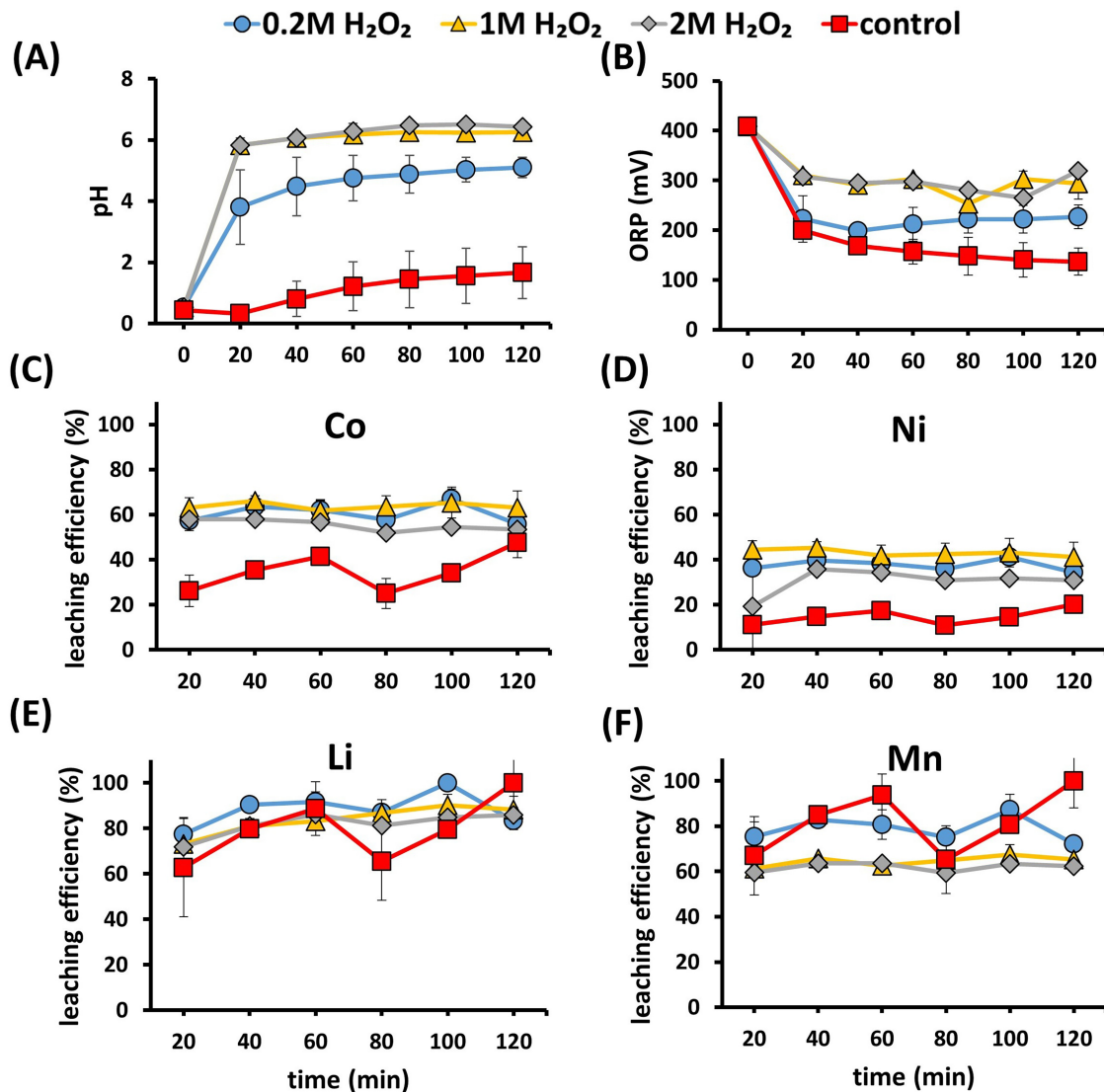


FIGURE 1

Changes in (A) pH, and (B) ORP, and leaching efficiencies of (C) Co, (D) Ni, (E) Li, and (F) Mn during indirect leaching of 10% (w/v) NMC using biogenic H_2SO_4 acid with different concentrations of H_2O_2 (0.2, 1, and 2 M). The error bars represent standard deviations ($n = 2$).

metal-biosurfactant aggregates, reducing thus the dissolution of Mn, Ni, and Co.

Rhamnolipids have been used for the removal of heavy metal contaminant such as Ni, Cu and Cd from manganese nodules (Lee and Kim, 2019). Biosurfactant-based emulsion liquid membrane proved to be an effective and cost-efficient treatment for the removal of Mn^{2+} from contaminated water, achieving more than 77% Mn removal in just 2 min (Ferreira et al., 2019). Tang et al. (2020) demonstrated that a combination of a biosurfactant and an electrokinetic technique enhances the removal of metals, including Ni, Mn, Cr, Pb, Zn and Cu, from sewage sludge. The biosurfactant surfactin has been successfully employed as a collector in ion flotation, resulting in the recovery of 94.7 % Ni and 98.2% Co from synthetic wastewater solutions (Schlebusch et al., 2023). Therefore, biosurfactant have a potential for selective recovery of critical metals such as Co, Ni and Mn from leachate solution produced during biohydrometallurgical treatment of spent LIB. Further investigation should focus on evaluating broader range of

biosurfactant and optimizing their concentrations to enhance metal recoveries and selectivity.

3.3 Scale up of H_2SO_4 – H_2O_2 system

The indirect leaching of NMC using biogenic H_2SO_4 and 0.2 M H_2O_2 was scaled up to a 1 L reactor (Figure 3). Compared to the flask experiments (Figure 1), leaching efficiencies of Li and Mn increased significantly, likely due to enhanced mixing and improved solid-liquid contact under reactor conditions. The addition of H_2O_2 improved Co and Ni dissolution by ~10% and ~15%, respectively, compared to a H_2O_2 -free control. In the H_2O_2 -amended system, leaching efficiencies reached 61.2% for Co and 32.8% for Ni after 1 h, increasing further to 67.5% for Co and 36.6% for Ni after 24 h of leaching (Figure 3C). These results are consistent with (Meshram et al. (2016), who reported an increase in Co recovery from 66.2% to 79.2% upon addition

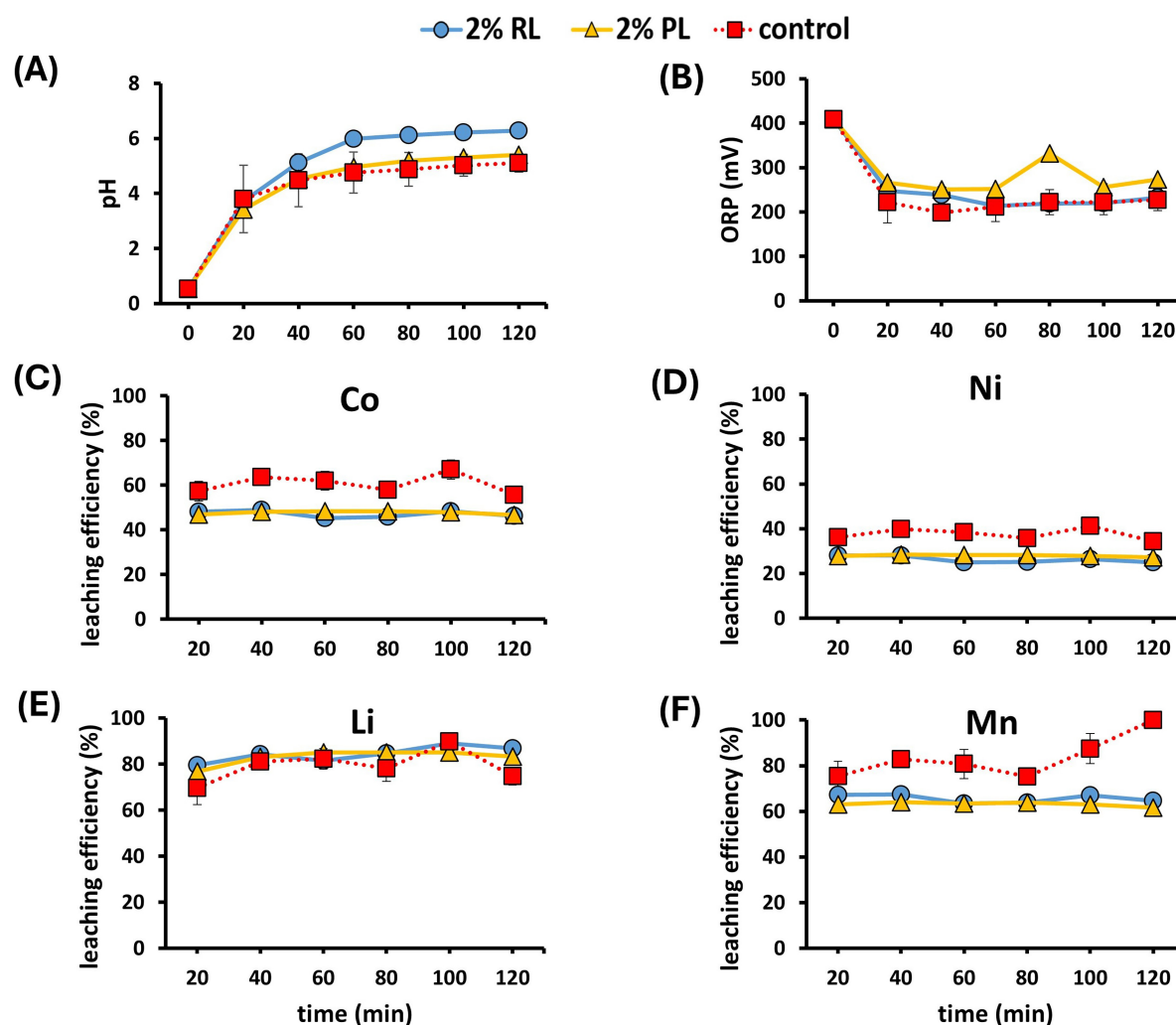


FIGURE 2

Changes in (A) pH (B) ORP, and leaching efficiencies of (C) Co, (D) Ni, (E) Li, and (F) Mn during indirect leaching of 10% (w/v) NMC using biogenic H_2SO_4 and 0.2 M H_2O_2 , with additions of 2% (v/v) biosurfactants rhamnolipids (RL) and phospholipids (PL) after 40 min of leaching. H_2SO_4 - 0.2 M H_2O_2 without biosurfactant is used as control. The error bars represent standard deviations ($n = 2$).

of 5% H_2O_2 during NMC leaching with 1 M H_2SO_4 . Miao et al. (2026) reported that dynamic dosing of H_2O_2 , rather than a single step addition (as employed in the present study), maintains reducing condition throughout the leaching process, facilitating the complete reduction of Ni^{3+} and Co^{3+} to their divalent states. Therefore, adopting such dosing strategy could further improve the leaching efficiency, particularly for Co and Ni. For Li and Mn, complete recovery was achieved within 60 min in both H_2O_2 -containing and H_2O_2 -free systems, further supporting that proton attack alone is sufficient to promote their dissolution (Figure 3C). Similarly, during a scale-up (1 L) two-step bioleaching of NMC-based BM, in which the effect of magnetic field was investigated, complete recovery of Mn and Li was also achieved but with relatively low pulp density of 3% (w/v) and after 3 days of leaching time (Hong et al., 2025). Panda et al. (2024) successfully scale-up direct bioleaching of BM using an adapted consortium in a 10 L reactor, resulting in metal recoveries of up to 80% Li, 91% Ni, 75% Co and 75% Mn. Nevertheless, these recoveries were achieved using a relatively low pulp density of only 0.5 % (w/v), which may limit the industrial applicability of the process.

SEM/EDS analysis of the solid residues collected after the reactor experiments confirmed complete removal of Mn from the solid phase, while residual Co, Ni, and Al were still detected in the post-leaching solids (Figure 4). In NMC structures, Mn is typically present as Mn^{3+} and Mn^{4+} , and its reduction-dissolution to Mn^{2+} generally requires strongly acidic and reducing conditions. However, Meshram et al. (2015) reported that at $\text{pH} > 3.2$ and under low redox potential, Mn^{4+} can be indirectly solubilized via the formation of an intermediate Mn_2O_3 phase. The leaching of Mn in this study may therefore proceed through a similar pathway, i.e., through the formation of an intermediate phase, resulting in the high recovery of Mn in both reactor systems.

It has been reported that Cu and Al present in NMC can positively influence leaching behavior by acting as reducing agents that improve the overall metal dissolution (Joulié et al., 2017). The NMC used in this study contained relatively high concentrations of both Cu (3.5%) and Al (5.26%). However, Cu concentration in the leachates remained below the detection limit, while $<40\%$ of Al was dissolved in both reactors (results not shown). The limited Cu dissolution might be attributed to the absence of

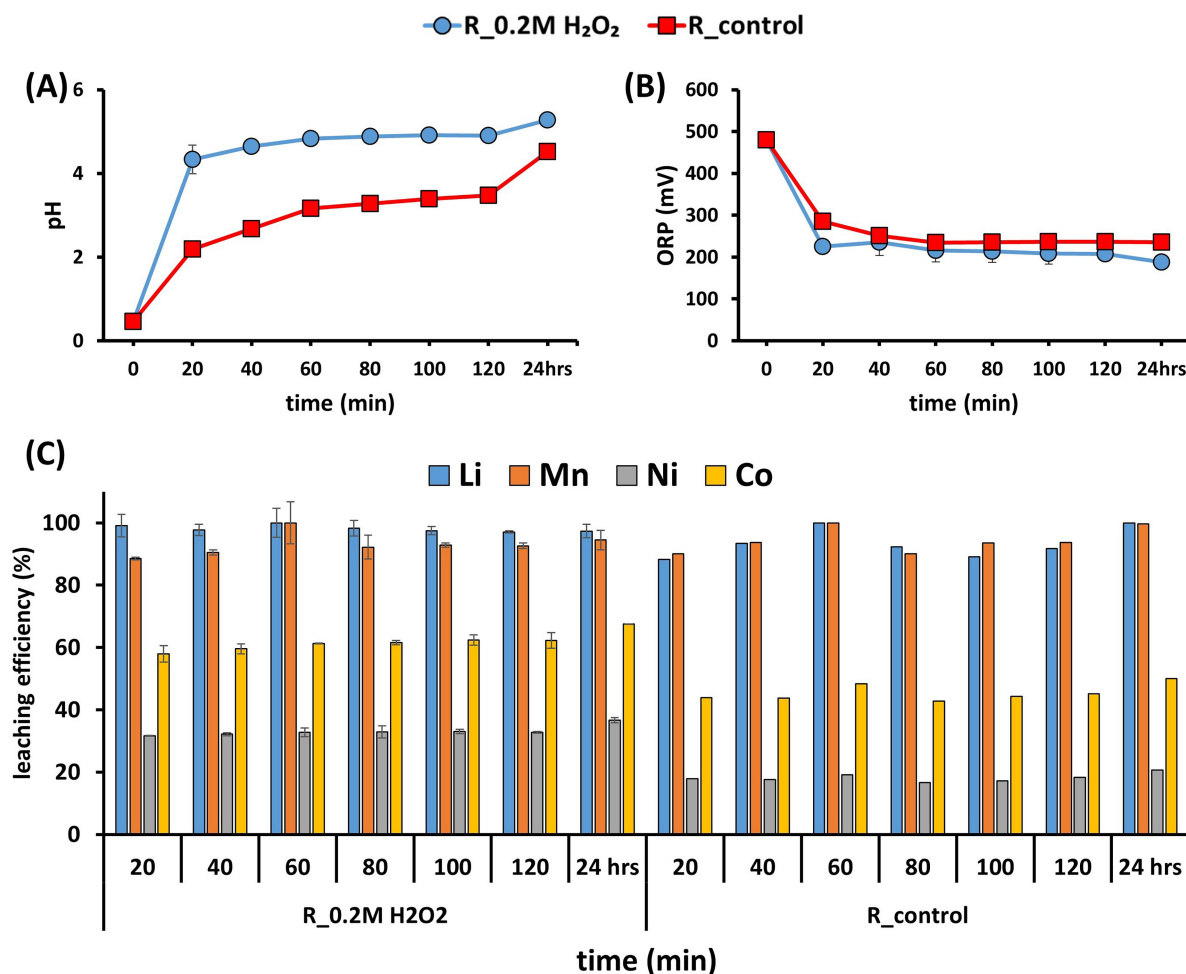


FIGURE 3 Changes in (A) pH and (B) ORP, and (C) leaching efficiencies of Li, Mn, Ni, and Co during upscaled indirect leaching of 10% (w/v) NMC using biogenic H₂SO₄ acid with 0.2 M H₂O₂ (R_0.2 M H₂O₂) without H₂O₂ (R_control). The error bars represent standard deviations ($n = 2$).

favorable redox cycling conditions (e.g., Fe²⁺/Fe³⁺ oxidation-reduction cycle), which are often associated with enhanced Cu mobilization. In contrast, Al behavior is highly pH-dependent, with precipitation expected to start occurring at pH 3–4, which may have limited its recovery under the conditions investigated. Therefore, optimization of the leaching conditions for both Cu and Al may further improve the recovery of Co and Ni (Porvali et al., 2020; Yan et al., 2023; Lalropuia et al., 2024).

Most previous studies on NMC leaching have been conducted at elevated temperatures (50–100 °C) and using concentrated acids (>1 M) (Baniyadi et al., 2025; Meshram et al., 2015; Mukai et al., 2024; Nadimi and Jalalian Karazmoudeh, 2020; Wang et al., 2016). In contrast, the present study operates at room temperature (22 °C) without external heating and uses dilute H₂SO₄ (0.6 M), thereby significantly reducing energy input and improving the potential economic viability of scale-up. The leaching agent, biogenic H₂SO₄, was produced from sulfur waste using chemolithotrophic bacteria that utilize CO₂ as the sole carbon source, thereby enhancing process sustainability and potentially contributing to carbon neutral pathway. High recoveries of strategic metals, particularly Mn and Li, were achieved at a high pulp density of 10% (w/v) within 1 h of leaching.

Overall, these findings demonstrate a scalable and sustainable bioleaching strategy for the efficient recovery of critical metals from NMC cathode waste. However, the incubation period for each cycle (14–21 days) is relatively long compared with conventional commercial acid production processes. Further optimization of the experimental condition is still necessary to reduce the incubation time and improve process efficiency. Välimets et al. (2026) reported that increasing the concentration of biogas derived sulfur substrate to 6% (w/v) resulted in an increase in acid concentration to as high as 2 M. The use of higher acid concentration employed in this study (0.6 M) could further enhance the kinetics of metal leaching from BM. Previous studies have demonstrated that bioleaching presents a cost-effective and environmentally friendly approach for metal recovery from different waste streams. Techno-economic analysis of LIB bioleaching using *Gluconobacter oxydans* demonstrated strong economic feasibility, achieving a positive net present value of \$136 million and an 11% profit margin for a scale-up plant, while enabling efficient recovery of critical metals with a lower carbon footprint compared to conventional recycling processes (Khan et al., 2025). A sensitivity analysis conducted during techno-economic assessment of tungsten recovery from semiconductor wastewater via bioleaching highlighted pulp density and microbial

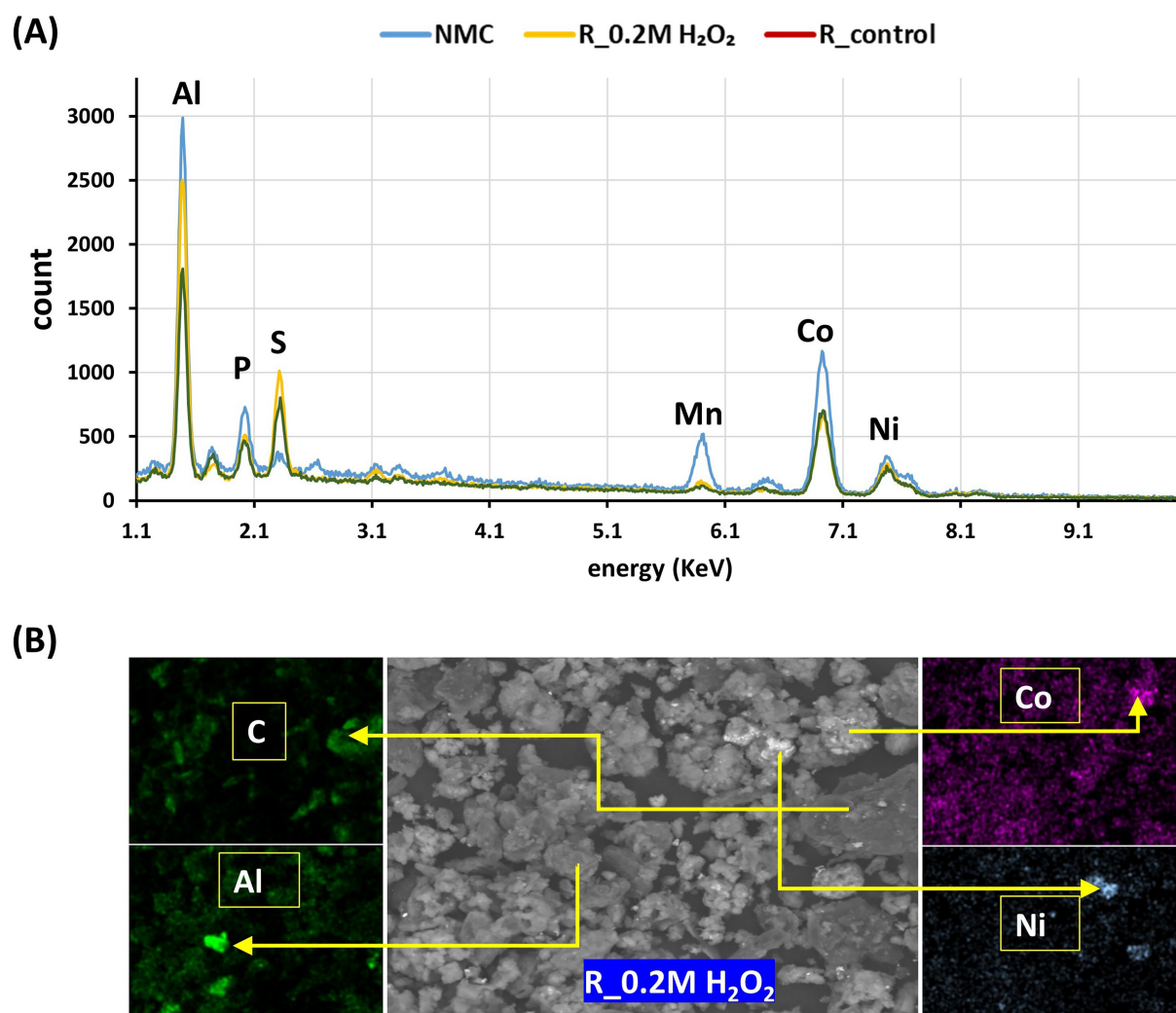


FIGURE 4

(A) SEM/EDS analysis of the original NMC-based black mass (NMC) and the solid residues collected after indirect leaching of 10% NMC using biogenic H₂SO₄ in a 1 L bioreactors, with (R_0.2 M H₂O₂) and without (control) 0.2 M H₂O₂; (B) SEM/EDS elemental mapping image (C, Al, Co, and Ni) of the solid residue collected from the reactor with 0.2 M H₂O₂.

cultivation time as key factor affecting the economic viability and scale-up feasibility (Lee et al., 2024). Furthermore, the current study does not include techno-economic analysis or life cycle analysis, both of which are essential for evaluating the economic feasibility and environmental sustainability of the proposed process and to enable comprehensive comparison with conventional LIB recycling technologies. Addressing these limitations in future studies could further improve the process efficiency, scalability, and practical applicability of biogenic acid-based LIB recycling.

4 Conclusion

Microbially produced H₂SO₄, using elemental sulfur recovered from a biogas plant, was successfully used for leaching black mass from NMC-based cathode material. The effect of varying H₂O₂ concentrations on metal solubilization was investigated, with 0.2 M identified as the optimum concentration for maximum

metal extraction efficiency. Biosurfactants (rhamnolipids and phospholipids) were also evaluated in combination with H₂O₂, however, no improvement in metal leaching was observed. Finally, the optimized H₂SO₄-H₂O₂ was upscaled in a 1 L reactor, achieving 100% recovery of Mn and Li from 10% (w/v) NMC within 1 h. These findings underscore the high potential of biogenic acid for efficient recovery of critical metals from spent LIB. However, despite its high potential, further assessment of the economic feasibility and life cycle impacts is still required to validate this process as a cost effective and sustainable circular economy-based alternative for LIB recycling.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Author contributions

LL: Investigation, Visualization, Conceptualization, Data curation, Writing – review & editing, Formal analysis, Methodology, Writing – original draft. KM: Formal analysis, Data curation, Investigation, Writing – review & editing. WR: Data curation, Writing – review & editing, Formal analysis. EP: Supervision, Conceptualization, Methodology, Writing – review & editing. GG: Project administration, Supervision, Writing – review & editing, Funding acquisition, Resources, Validation.

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